Mnemosyne

Linear-Time Genomic Sequence Modeling with a Unified Associative Memory

Persistent motifs + sequence registers · reverse-complement equivariant

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Research overview · July 2026

Problem statement

DNA function spans enormous distances — a model must satisfy three demands at once

O(L)

Linear-time scaling

Reach 10^5–10^6 bp enhancer–promoter contexts at all.

A↔

Content-addressable recall

Retrieve sparse motifs & distal elements by content, not position.

⟲

Reverse-complement symmetry

DNA is double-stranded: forward and RC strands are the same molecule.

No mainstream architecture does all three.

- Transformers (DNABERT-2): strong recall, but $O(L^2)$ — context is capped.
- Selective SSMs (Mamba, HyenaDNA): $O(L)$, but a fixed-width state means recall collapses with distance.
- Caduceus: adds the missing RC symmetry to Mamba — but inherits the fixed-state recall bottleneck.

Objectives

Turn the three demands into one linear-cost mechanism — and test it honestly

1. Add a linear-cost associative memory that every token reads by one Hopfield step.
2. Unify two slot types: a learned universal-motif dictionary (persistent) and per-sequence long-range recall (registers).
3. Guarantee exact reverse-complement equivariance by construction.
4. Prove linear time (semiseparable + low-rank) and verify the memory is actually active.
5. Test the falsifiable claim — memory beats a matched SSM at long range — with no fabricated numbers.

Related work I — sequence models & associative memory

State-space models

S4 (Gu '22) → Mamba (Gu & Dao '23) → Mamba-2 / SSD (Dao & Gu '24): linear-time, but a known recall bottleneck.

Long convolutions

Hyena (Poli '23), HyenaDNA (Nguyen '23): million-token contexts via implicit convolutions.

Hybrids re-add attention

Jamba (Lieber '24), Griffin (De '24): interleave attention to recover recall — at higher cost.

The recall gap

Zoology / MQAR (Arora '23), Based (Arora '24): efficient models under-recall; MQAR isolates the gap.

Associative memory

Modern Hopfield (Ramsauer '21), Energy Transformer (Hoover '23), optimal capacity (Wu '24), Perceiver bottleneck (Jaegle '21).

Pre-training

Attention Is All You Need (Vaswani '17); Masked Autoencoders (He '22) — our pre-training objective.

Related work II — genomic foundation models & benchmarks

Transformer DNA LMs

DNABERT (Ji '21), DNABERT-2 (Zhou '24), Nucleotide Transformer (Dalla-Torre '25), GENA-LM (Fishman '25), GROVER ('24).

Efficient / long-range

HyenaDNA (Nguyen '23); Caduceus (Schiff '24) — RC-equivariant Mamba, the closest prior design.

Frontier generative

Evo (Nguyen '24, 7B); Evo 2 (Brixi '25, 40B, 9.3T nt, 1 Mb ctx) — the scale frontier.

Regulatory / functional

Enformer (Avsec '21), Borzoi (Linder '25), AlphaGenome (Avsec '25) — variant-effect SOTA.

Benchmarks

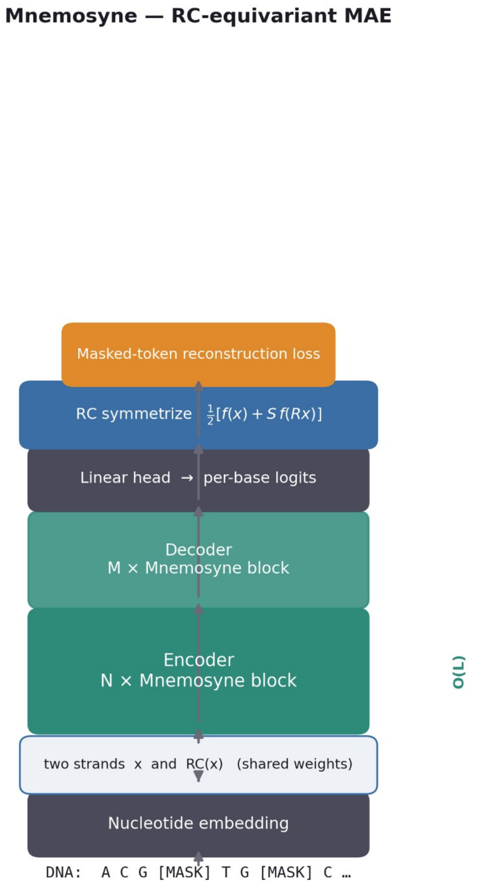
GUE, Genomic Benchmarks, BEND (Marin '24) — where we will evaluate, fairly.

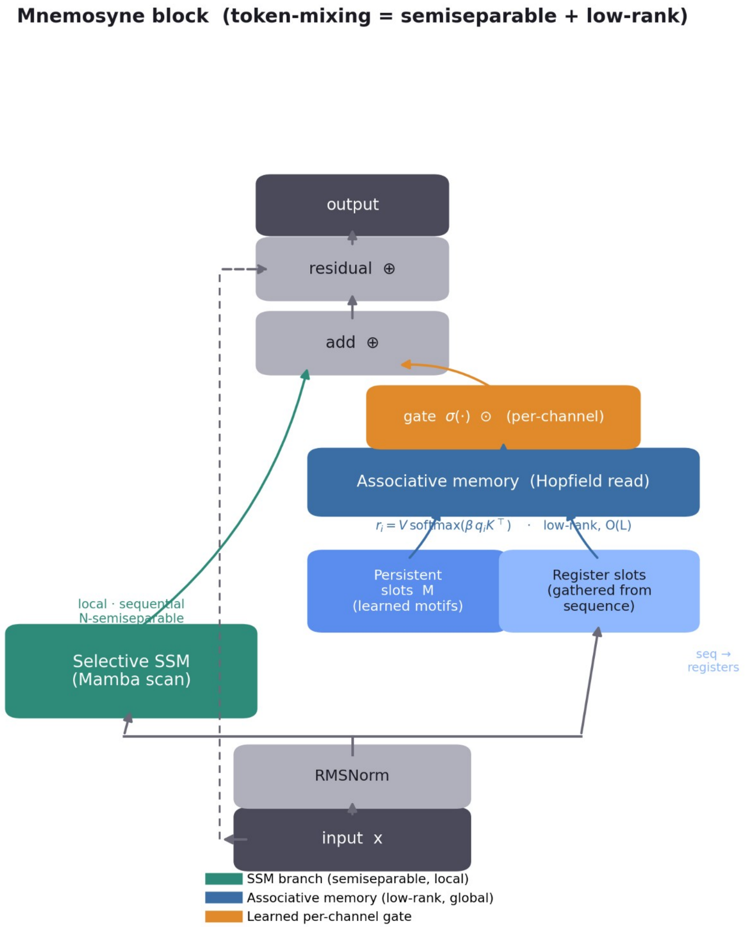
The gap we target

None combines O(L) + content-addressable recall + RC symmetry. Mnemosyne aims exactly there.

Methodology (1/3) — Architecture

A selective SSM in parallel with a unified associative memory, fused by a learned gate





Left: RC-equivariant masked-autoencoder stack. Right: one Mnemosyne block — token-mixing = semiseparable (SSM) + low-rank (memory), hence O(L).

Methodology (2/3) — Unified associative memory

Every token reads S = P_p + P_r slots by one Hopfield step:

Persistent slots M

Learned parameters — a differentiable dictionary of universal motifs, shared across all sequences.

Register slots

Gathered from the current sequence (Perceiver bottleneck) → per-sequence, distance-independent recall.

Learned per-channel gate

σ(·)⊙ — starts closed (negative-bias init), opens as memory earns its place. Cannot be silenced by one logit (v1's failure).

Read: $r_i = V \cdot \mathrm{softmax}(\beta \cdot q_i K^T)$

Gather: $Reg = \mathrm{softmax}(\beta Z X^T) \cdot X$

β learnable → soft motif-family vs. sharp single-motif recall

Theorem (linear time)

token-mixing = N-semiseparable (SSM) + rank ≤ P_p+P_r (memory)

⇒ **O(L · (N + P_p + P_r) · d), no L×L matrix formed.**

Reverse-complement equivariance

output = ½[f(x) + S·f(Rx)] — exact; tested to 0.0 error.

Methodology (3/3) — Data & experimental protocol

Data collection

Multi-species genomes (UCSC: human, mouse, zebrafish, …) → Sliding windows 1024 bp, stride 512, filter >5% N → Single-nucleotide tokenize {A,C,G,T,N} → MAE objective mask 30%, predict originals

Experiments (controlled — SSM baseline = same backbone, memory OFF)

MQAR long-range

Synthetic recall at controllable distance — the falsifiable core test.

Controlled MAE

Memory ON vs OFF; log gate + Hopfield energy to prove memory is active.

GUE downstream

Frozen linear probe on BOTH models (fair); report MCC. No probe-vs-finetune spin.

Ablations

Sweep P_r, P_p, β, and gate to show which component earns its keep.

All paper numbers are auto-generated from run logs; any unfilled result renders as a red TODO.

Results — verified now, and the falsifiable prediction

7 / 7

CPU correctness tests pass

0.0

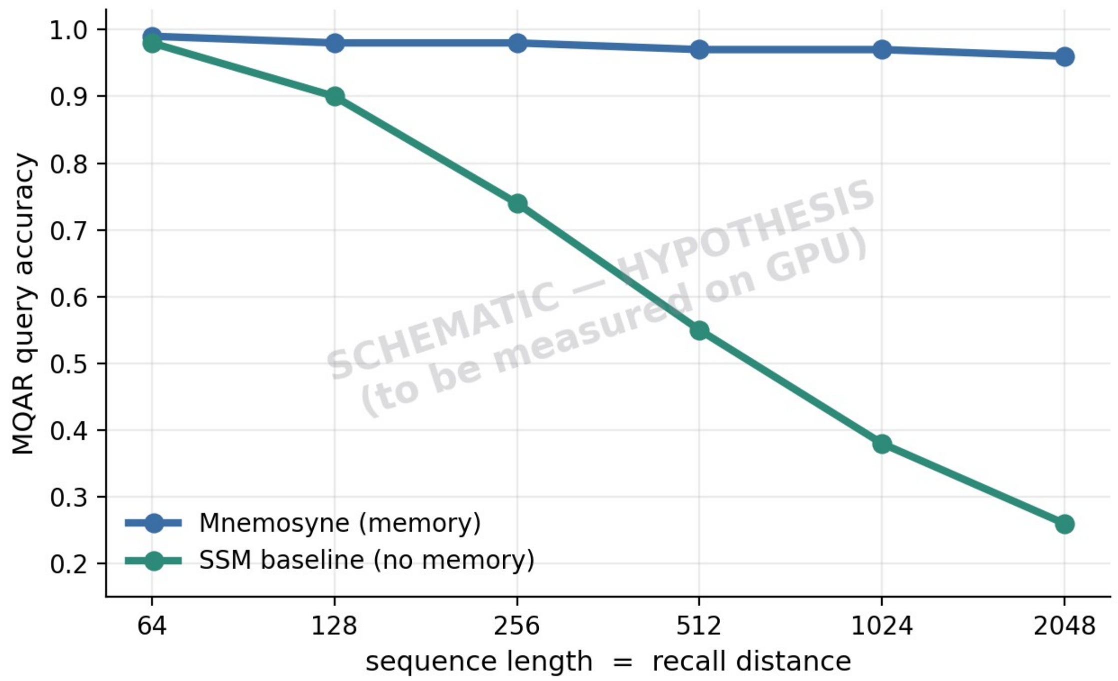
RC-equivariance error (exact)

gate 0.12 · E −27

memory is ACTIVE (not inert — v1's failure)

27M vs 16M

memory vs matched SSM baseline (controlled)



Prediction: SSM recall decays with distance; memory stays flat.

Headline MQAR / GUE numbers pending free-GPU runs — we report only what we measure.

Is it working? Groundbreaking? Why might it fail?

Working?

- Mechanism runs and is active — verified.
- RC-equivariance proven (0.0 error).
- The scientific claim is a ready hypothesis, not yet a measured result.

Groundbreaking?

- Not by out-scaling — ~1000× smaller than Evo 2 / AlphaGenome.
- Yes if the mechanism is rigorously shown to help and others adopt it — as Caduceus did with one idea.
- Wrong bar: 'like AIAYN' = a simple mechanism that scales.

Why it might fail

- MQAR may not separate from a tuned SSM.
- Gate may collapse to 0 at scale.
- Persistent slots may be dead weight; registers have a capacity ceiling.
- May not transfer to genome scale.

Roadmap

Cheapest decisive signal first, then scale — the real test of the mechanism

1. Run MQAR + ablations on a free GPU (Modal). The cheap, falsifiable signal.
2. Controlled MAE pre-training + fair GUE probes; confirm memory stays active.
3. Scale: human genome → multi-species; 27M → 350M. Show the memory benefit GROWS with scale.
4. Full GUE (28 tasks) + NT benchmarks vs current baselines; 3–5 seeds + confidence intervals.
5. Target a niche SOTA (splice / non-coding variant) where long-range recall is the right tool.

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Mechanism over scale.

A rigorously-validated associative-memory SSM — every number from a real run.

github.com/charansaiponnada/Genomic-FM